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Java SE 27 & JDK 27

Class Dictionary<K,V>

java.lang.Object
    java.util.Dictionary<K,V>

Type Parameters:

K - the type of keys

V - the type of mapped values

Direct Known Subclasses:

Hashtable

```
public abstract class Dictionary<K,V>
extends Object
```

The Dictionary class is the abstract parent of any class, such as Hashtable, which maps keys to values. Every key and every value is an object. In any one Dictionary object, every key is associated with at most one value. Given a Dictionary and a key, the associated element can be looked up. Any non-null object can be used as a key and as a value.

As a rule, the equals method should be used by implementations of this class to decide if two keys are the same.

NOTE: This class is obsolete. New implementations should implement the Map interface, rather than extending this class.

Since:

1.0

See Also:

Map,
Object.equals(java.lang.Object),
Object.hashCode(),
Hashtable

Constructor Summary

Constructors

Constructor	Description
Dictionary()	Sole constructor.

Method Summary

All Methods | **Instance Methods**

Modifier and Type	Method	Description
abstract Enumeration<V>	**elements**()	Returns an enumeration of the values in this dictionary.
abstract V	**get**(Object key)	Returns the value to which the key is mapped in this dictionary.
abstract boolean	**isEmpty**()	Tests if this dictionary maps no keys to value.
abstract Enumeration<K>	**keys**()	Returns an enumeration of the keys in this dictionary.
abstract V	**put**(K key, V value)	Maps the specified key to the specified value in this dictionary.
abstract V	**remove**(Object key)	Removes the key (and its corresponding value) from this dictionary.
abstract int	**size**()	Returns the number of entries (distinct keys) in this dictionary.

Methods declared in class Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Constructor Details

Dictionary

```
public Dictionary()
```

Sole constructor. (For invocation by subclass constructors, typically implicit.)

Method Details

size

```
public abstract int size()
```

Returns the number of entries (distinct keys) in this dictionary.

Returns:

the number of keys in this dictionary.

isEmpty

```
public abstract boolean isEmpty()
```

Tests if this dictionary maps no keys to value. The general contract for the `isEmpty` method is that the result is true if and only if this dictionary contains no entries.

Returns:

`true` if this dictionary maps no keys to values; `false` otherwise.

keys

```
public abstract Enumeration<K> keys()
```

Returns an enumeration of the keys in this dictionary. The general contract for the keys method is that an `Enumeration` object is returned that will generate all the keys for which this dictionary contains entries.

Returns:

an enumeration of the keys in this dictionary.

See Also:

`elements()`, `Enumeration`

elements

```
public abstract Enumeration<V> elements()
```

Returns an enumeration of the values in this dictionary. The general contract for the `elements` method is that an `Enumeration` is returned that will generate all the elements contained in entries in this dictionary.

Returns:

an enumeration of the values in this dictionary.

See Also:

`keys()`, `Enumeration`

get

```
public abstract V get(Object key)
```

Returns the value to which the key is mapped in this dictionary. The general contract for the `isEmpty` method is that if this dictionary contains an entry for the specified key, the associated value is returned; otherwise, `null` is returned.

Parameters:

`key` - a key in this dictionary. `null` if the key is not mapped to any value in this dictionary.

Returns:

the value to which the key is mapped in this dictionary;

Throws:

`NullPointerException` - if the `key` is `null`.

See Also:

`put(java.lang.Object, java.lang.Object)`

put

```
public abstract V put(K key,
                      V value)
```

Maps the specified `key` to the specified `value` in this dictionary. Neither the key nor the value can be `null`.

If this dictionary already contains an entry for the specified `key`, the value already in this dictionary for that `key` is returned, after modifying the entry to contain the new element.

If this dictionary does not already have an entry for the specified `key`, an entry is created for the specified `key` and `value`, and `null` is returned.

The `value` can be retrieved by calling the `get` method with a `key` that is equal to the original `key`.

Parameters:

`key` - the hashtable key.

`value` - the value.

Returns:

the previous value to which the `key` was mapped in this dictionary, or `null` if the key did not have a previous mapping.

Throws:

NullPointerException - if the key or value is null.

See Also:

Object.equals(java.lang.Object),
get(java.lang.Object)

remove

```
public abstract V remove(Object key)
```

Removes the key (and its corresponding value) from this dictionary. This method does nothing if the key is not in this dictionary.

Parameters:

key - the key that needs to be removed.

Returns:

the value to which the key had been mapped in this dictionary, or null if the key did not have a mapping.

Throws:

NullPointerException - if key is null.

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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